

08 | OVERVIEW



Evaluating Scientific Confidence: A Case Study of Climate Reports and Peer Review

Abstract

In late 2012, a draft of the Intergovernmental Panel on Climate Change's fifth assessment report was leaked online before review was complete. Part of it was quickly misrepresented. This lesson uses the incident to explore how scientists express confidence, why peer review matters, and how taking a sentence out of context can distort what science actually says.

Introduction

The Intergovernmental Panel on Climate Change (IPCC) brings together scientists from around the world to assess published climate research. Its reports go through multiple rounds of review before they are finalized. In December 2012, a blogger who had signed up as a reviewer posted a draft of the fifth assessment report online, breaking a pledge of confidentiality. The draft stated that it was virtually certain that human activities, primarily rising carbon dioxide, were causing an imbalance in Earth's energy budget. Yet the blogger highlighted a different section about cosmic rays, claiming it changed everything. Scientists responded by explaining what the draft really said, and why unfinished reports should stay confidential.

How Do Scientists Study This?

IPCC reports are assessments, not new experiments. Author teams review the published, peer-reviewed literature and summarize what it shows, using calibrated language to express how certain they are. Drafts are sent to expert and government reviewers, whose comments must be considered before the report is approved. In this case study, scientists examined the leaked draft and compared the blogger's claim with the full text. They also compared the draft's confidence language with the previous report from 2007 to see how scientific certainty about human-caused warming had changed over time.

How an IPCC report is made



Assess the evidence

Author teams summarize published, peer-reviewed research rather than running new studies.

Review the draft

Experts and governments comment on drafts, and authors must respond before the report is final.

State the confidence

Calibrated terms such as virtually certain tell readers how strong the evidence is.

What Did They Discover?

The leaked draft stated there was consistent evidence that Earth was taking up more energy than it was losing, and that this was virtually certainly caused by human activities. Natural factors, it said with very high confidence, contributed only a small fraction. Psychologist Stephan Lewandowsky explained that virtually certain meant 99% confidence, up from 90% (very high confidence) in the 2007 report. That level of confidence, he said, is comparable to our confidence in basic laws of physics such as gravity. Report author Steve Sherwood explained that the cosmic ray section had been taken out of context and could not be used to cast doubt on human-caused warming.

Confidence in human-caused warming

90%

very high confidence in the 2007 IPCC report

99%

virtually certain in the leaked 2012 draft

5th

assessment report cycle for the IPCC

Lewandowsky noted that science is highly transparent, but drafts must stay confidential until they are checked through peer review.

Takeaways

Certainty grew

Confidence that humans cause warming rose from 90% in 2007 to 99% in the draft.

Context matters

Quoting part of a report out of context can reverse its actual meaning.

Review improves science

Drafts are unfinished; peer review makes the final report more accurate and complete.

Key Terms

Peer review

The process in which independent experts check scientific work for accuracy and quality before it is finalized or published.

Confidence language

Standard terms, such as very likely or virtually certain, that scientists use to express how strong the evidence is.

Energy imbalance

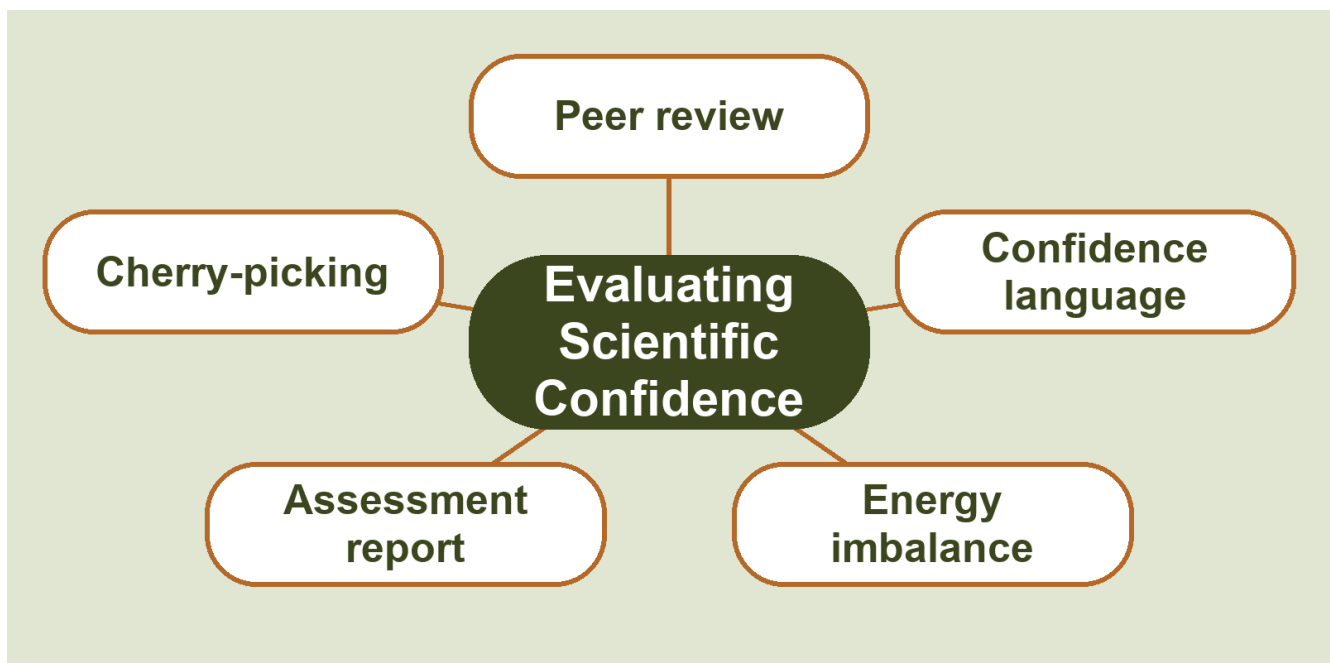
A difference between the energy Earth absorbs from the Sun and the energy it releases back to space, which causes warming.

Assessment report

A summary of existing research on a topic, written to inform decision-makers.

Cherry-picking

Selecting only the pieces of evidence that support a claim while ignoring evidence that contradicts it.



08 | ACTIVITY

Activity

Out of Context

Students practice spotting how a quotation can be distorted by removing its context.

Materials: A short science paragraph provided by the teacher; Highlighters; Confidence scale handout: likely, very likely, virtually certain

1. Place the terms very high confidence (90%) and virtually certain (99%) on a confidence scale.
2. Read the paragraph and highlight its main conclusion.
3. Pick one sentence that, taken alone, could make it seem to say something different. Write a misleading headline using only that sentence.
4. Swap headlines with another group and use the full paragraph to correct theirs.
5. Discuss why reviewers agree to keep drafts confidential until peer review is complete.

References

Creagh, S. (2012). Human role in climate change now virtually certain: leaked IPCC report. The Conversation, with comments from Kevin Trenberth, Stephan Lewandowsky, and Steven Sherwood. <https://doi.org/10.64628/AA.y45975jwk>